SUB-PBA & CHARGING IC (BQ25890) SCHEMATIC

USB Type-C & Battery Fast Charging Architecture | Sheet 1 of 2

USB TYPE-C FRONT END & OVP PROTECTION

Over-Voltage Protection & VBUS Distribution

VBUS Input: 5V / 9V / 12V QC/PD fast charge input | Diode: 580mV

OVP FET (WS3222D): Clamps VBUS if voltage exceeds 14.5V within 50ns

CC1 / CC2 (Configuration Channels): 5.1K pull-down for 5V 3A detection

D+ / D- High-Speed Lines: Fast charge handshake (QC 3.0 / AFC / FCP)

VBUS to PMIC Input Capacitor: 10uF 25V X5R ceramic cap (frequent short point)

Thermal Shutdown Threshold: 145 deg C with automatic recovery at 120 deg C

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USB Type-C & Battery Fast Charging Architecture | Sheet 2 of 2

BQ25890 SWITCH-MODE BUCK CHARGER

High-Efficiency 5A Single-Cell Charger Stage

SW Pin (Switching Node): 1.5MHz high frequency square wave to coil L3001

SYS Pin (System Output): Automatically arbitrated between VBUS and Battery

BAT Pin: Direct Kelvin-sense line to Li-Ion battery pack (Charge cut-off: 4.40V)

TS (Temperature Sense): Connects to 10K NTC thermistor inside battery flex

STAT Pin (Open-Drain): Low during active charge, High-Z when charge complete

I2C Control Bus: 400KHz fast mode to CPU for programmable charge currents

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